

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Outperform Optical and ImageNet Transfer Across ViT and CNN?

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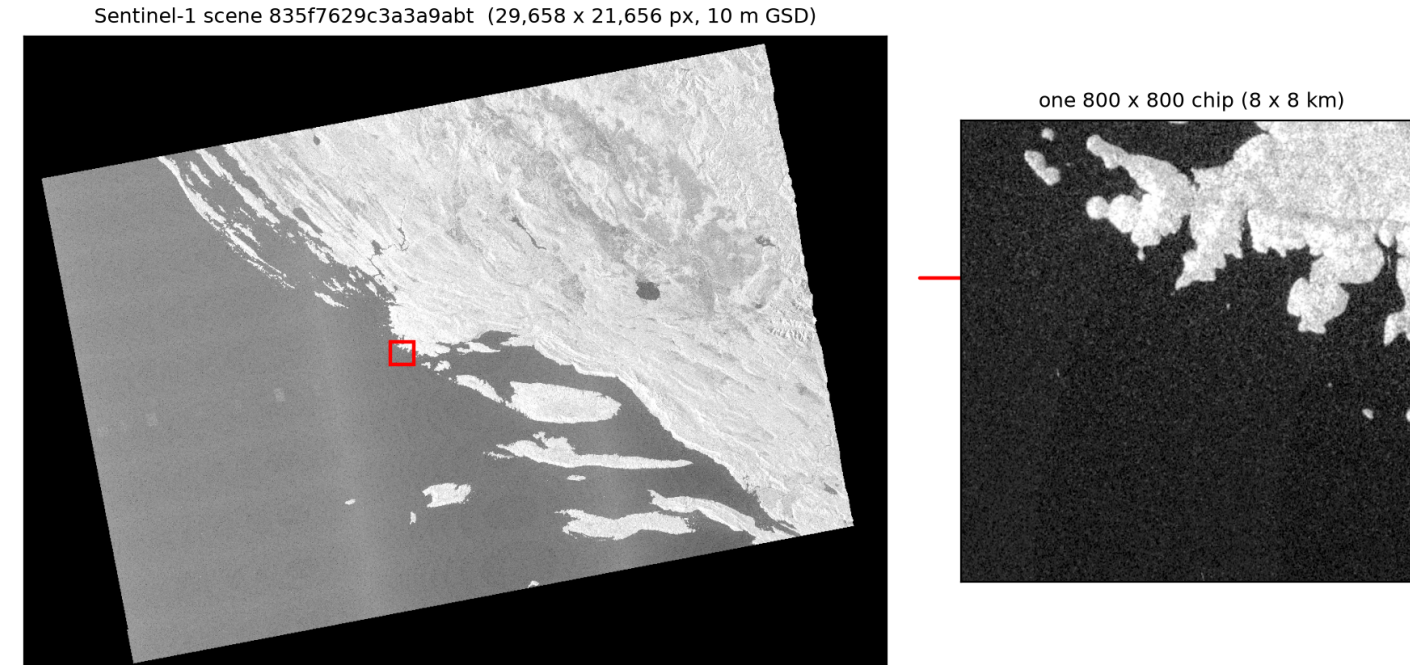
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Short answer: not consistently. With 12 labeled scenes every pretrained encoder beats random initialization, but SAR pretraining beats optical only for the CNN, and only while labels are scarce.

Motivation

- **Dark vessels** have no matching AIS report. Sentinel-1 SAR [2] sees them at night and through cloud.
- Vessels fill a few pixels in a 25,000-pixel scene. Sea clutter and coastlines give strong false returns.
- **Human-verified labels are scarce.** Pretraining can lower the label cost, but which source domain helps most, and does the answer depend on the backbone?

Data: xView3-SAR [1]



One Sentinel-1 scene (about 300 × 220 km) and one 800-pixel chip (8 × 8 km).

- 150 frozen study scenes: **111 train, 23 dev, 16 test**.
- Nested budgets: **12 ⊂ 28 ⊂ 56 ⊂ 111** training scenes (10, 25, 50, 100%).
- **50 near-shore, human-verified scenes**, opened once after test scoring: 8,642 vessels, 3,644 dark, 2,686 within 2 km of shore.
- Input [VH, VV, VH–VV] in dB, 10 m pixels.

References

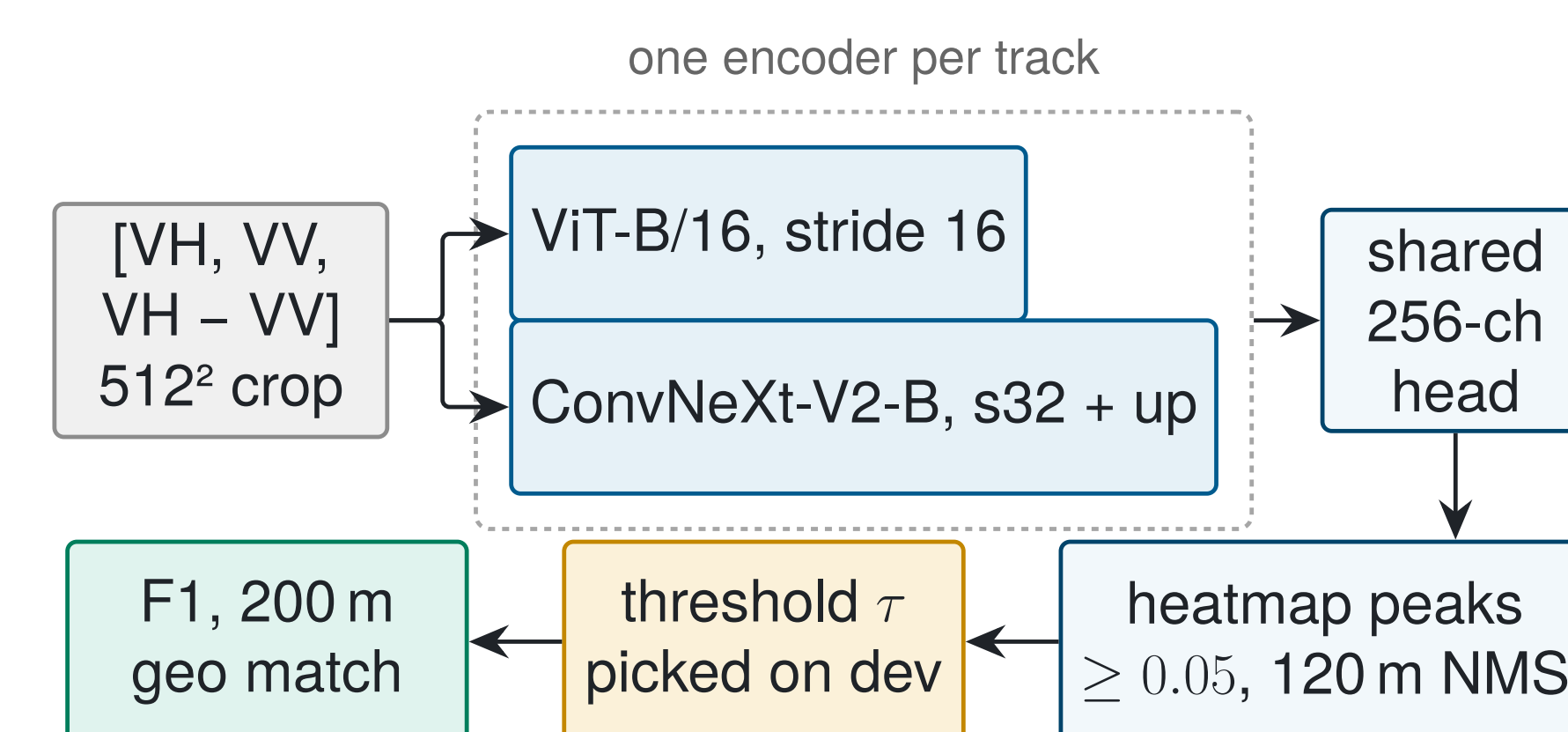
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8 arms × 4 budgets = 32 cells

Role	ViT-B/16 [3]	ConvNeXt V2 [4]
Random	fresh init	fresh init
Optical	SatDINO [6] (fMoW-RGB)	BigEarthNet-S2 [8]
SAR	SARMAE [7] (SAR-1M)	BigEarthNet-S1 [8]
ImageNet	AugReg [5]	FCMAE [4]

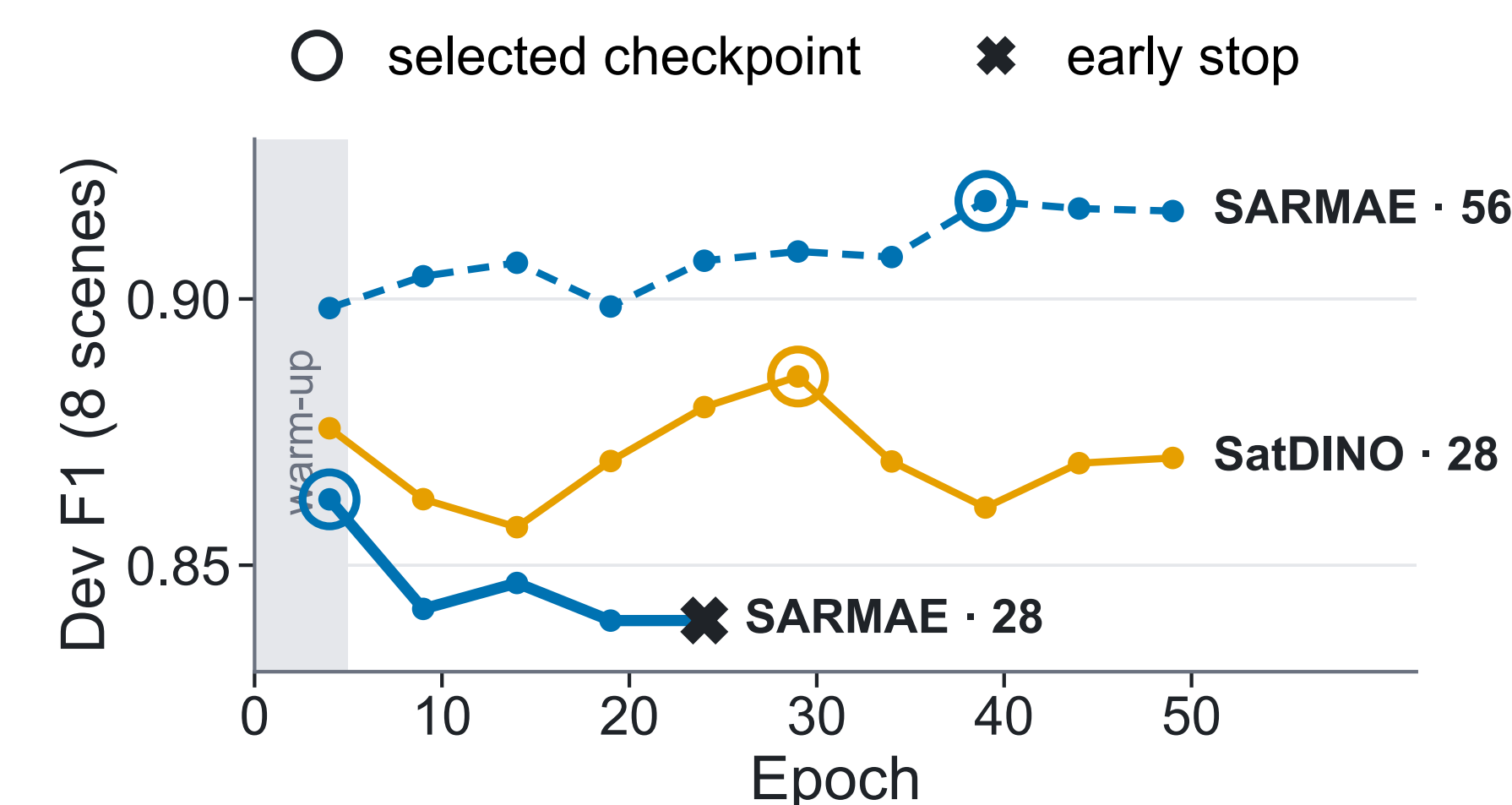
Only the initialization changes inside a track; we compare domains **within** an architecture.

One shared point detector



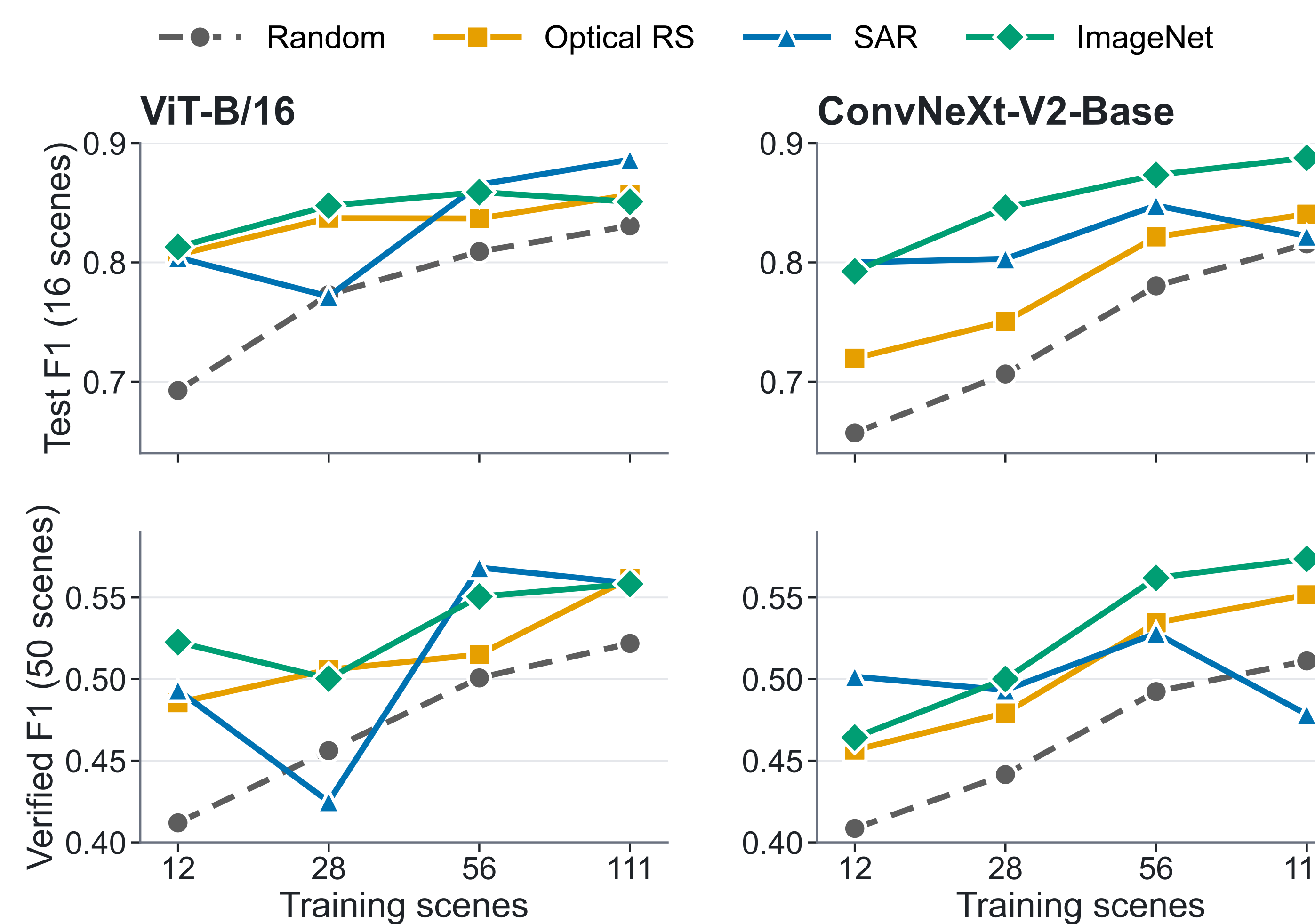
- CenterNet-style heatmap [9] with focal loss [10]; F1 with 200 m geographic matching.
- Same optimizer, schedule, crops, data order and seed (0) for every cell. Strict FP32: 567.6 H100 GPU-hours.
- 8 dev scenes pick each checkpoint and threshold; test and verified scenes are scored **once**, with no retuning.

Why SARMAE dips at 28 scenes



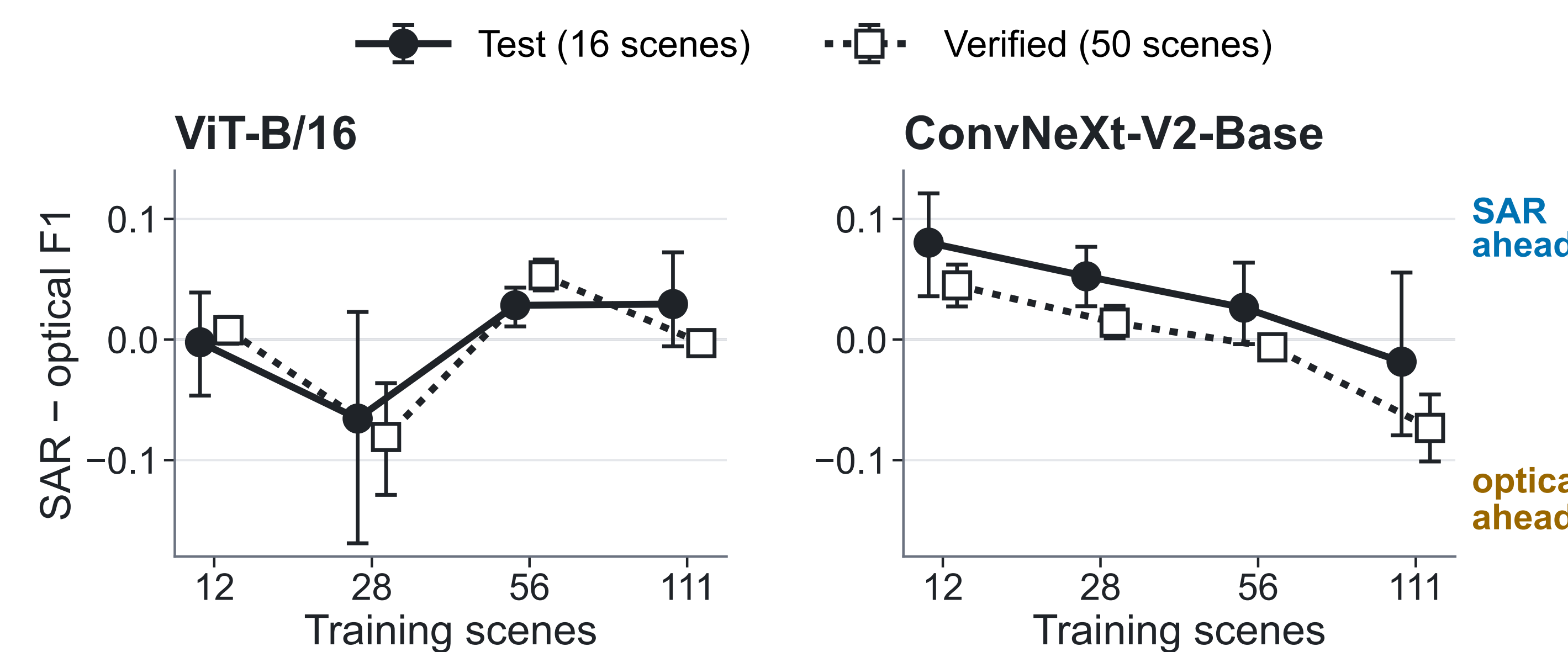
At 28 scenes SARMAE peaks as warm-up ends, then overfits (training loss 0.482 → 0.034; dev precision 0.858 → 0.768). Early stopping keeps the epoch-4 checkpoint.

Pretraining helps most when labels are scarce



F1 with 200 m matching. Each of the 32 cells is scored once with the threshold it picked on 8 dev scenes. Top: 16 held-out test scenes. Bottom: 50 near-shore, human-verified scenes. Seed 0.

SAR vs. optical depends on backbone and budget



Bars: 95% intervals from 10,000 paired scene resamples. They cover scene sampling only: retraining 28 unchanged cells moved test F1 by up to 0.043.

The verified scenes expose a coastal label gap

0.4% → 31% 34 / 4,302 0.724

vessels within 2 km of shore: training labels → verified scenes best cell's predictions within 2 km of shore best cell's offshore verified F1 (0.574 overall)

Every cell reaches at most 0.021 near-shore recall and 0.228 dark-vessel recall, while precision stays at 0.639–0.909: the detectors learned to stay silent near shore.

Findings

12 / 12

pretrained-vs-random comparisons at 12 training scenes have a 95% interval above zero.

- 1 Pretraining pays off with few labels.** Gains at 12 scenes: +0.063 to +0.143 test F1, +0.048 to +0.111 verified F1.
- 2 SAR beats optical only for the CNN, and only with few labels:** +0.080 [+0.036, +0.121] test F1 at 12 scenes, but −0.073 [−0.101, −0.046] verified F1 at 111. ViT: no stable order.
- 3 The input adapter matters.** With a seeded 3-channel stem instead of band slicing, BigEarthNet-S2 beats random at every budget (+0.025 to +0.063 test F1).
- 4 The coastal gap is a label gap.** Only 0.4% of training vessels lie within 2 km of shore, against 31% of verified ones.

Limits

- One seed per cell; 8 dev scenes pick checkpoints and thresholds.
- Monotonicity gate failed: two SAR test curves drop by more than 0.02 as labels grow.
- Checkpoints differ in objective and corpus, not only domain.

Open questions & next steps

- How do geospatial foundation models pre-trained jointly on SAR and optical imagery fare against these single-source checkpoints?
- Does SAR pretraining help where optical cannot: near shore and for small dark vessels?
- Would per-arm learning rates or longer early-stopping patience remove the small-budget dips?
- **Next:** near-shore training labels, a shoreline input channel, several seeds, and matched pretraining objectives across sensors.